

started out, like I said, knowing nothing. So we all had fun, since we were all playing at the same level.

But it's funny, we thought we were real bridge players, but we never could've gotten around and competed with real bridge players. A few years later when I was working at Hewlett-Packard, I tried to join a bridge club in my apartment building and I couldn't even begin to play with those women. You see, I'd never really memorized all those rules of how much you bid when you have which hands. So all I'd end up doing is messing up my partner.

These days, I can play bridge pretty well, but it's only because I read the bridge column in the newspaper every single day for years until I could figure out the formulas in my head.

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During college, I worked on one of my favorite projects ever. I called it the TV Jammer.

The TV Jammer came out of this thing I'd seen my old friend Allen Baum's father, Elmer, do over the summer. Mr. Baum was an engineer, and he'd worked out this little circuit on a piece of paper. It included a transistor, a couple of resistors, a capacitor, and a coil that could put out a signal in the TV frequency range. I looked at it, thinking how cool it would be if you could tune it, the same way you could tune your transistor radio, just by turning a dial. So I built a few of these—devices that let you jam a TV if you just dialed into the right frequency. They were cool.

Well, at some point during my freshman year at Colorado, I thought it was time to have fun with the TV Jammer. I walked over to Radio Shack and looked at all their transistors. I saw they had only one transistor rated for 50 MHz up toward the TV frequencies. I brought that one home. I also bought a little transistor radio so I could use parts of it, like resistors of certain values and the tuning capacitor, the part the tuning knob connects to. That would give me a big wide tuning range.

I hand-wrapped a coil out of some thick wire I had—about three turns—and I soldered on a little tap halfway down one of the turns and put a capacitor there. The whole thing was as small as my little finger, just tiny. I built it on top of the case of a 9-volt battery in a neat way. You know that little clip on top of the 9-volt? I stripped it out, hand-soldered it to the connectors on my little TV Jammer circuit, and then I could plug another 9-volt battery in as my power source. So I was able to carry this 9-volt battery case with the TV Jammer on it totally concealed. Except for a little six-inch wire that acted as an antenna, which I had to hang out the side to transmit. I put it up my sleeve to hide it.

I went over to a friend's to try it out on his TV—he had a little black-and-white TV in his dorm room—and sure enough, I was able to jam his TV black.

I walked into the main lounge of our dorm where everybody was watching a big black-and-white TV. I tuned the TV Jammer and, whack, it blacks out. Wow, I thought, that's a funny joke.

I showed it one day to Randy Adair, my Christian friend, and he said, "You should try it on the color TV that's in the basement of Libby Hall," the girls' dorm.

I walked in there and saw a lot of guys and girls. They were in there watching that TV all the time, it turned out. I walked in back where I was in the dark enough, and I turned the TV Jammer on, expecting it would kill the picture. All it did was fuzz it up, though.

Well, without any planning whatsoever, my friend Randy, sat in the front row of chairs, leaned over the TV, and whacked it really hard. I caught on quickly. I instantly made the TV picture go clear, which of course made everyone think that the whacking worked on the TV. I waited for a couple of minutes and jammed it again. It fuzzed up the picture again and Randy hit it again. And I made it go clear again. A couple of minutes later I jammed it

again, but this time I let Randy hit it three or four times before his whacking "worked."

So anyone watching would think that, okay, hitting harder works better. They all thought something was loose inside the TV and that by hitting it hard with your hand you could fix it. It was almost like a psychology experiment—except, I noticed, humans learn better than rats. Only the rats learn it quicker.

Then, later that night, Randy didn't get up to whack the TV. So someone else did. I was hoping that would happen! Someone else whacked it, and I made it so the TV worked. Ha! A whole audience of guinea pigs. I couldn't have wished for more. Over a period of about two weeks, I went there every night to watch people whack the TV. When that didn't work, they'd start to fine-tune it—in those days, TVs came with tuning controls—and I would quietly work the TV Jammer so that if they tuned it just right, the TV worked again.

After a while, I made it so that if someone touched the tuner and adjusted it to fix the picture, it would work. But then when they pulled their hand away, the screen would go bad again. Until they put their hand back on the tuning control, that is. I was like an entertainer. A puppeteer—with live puppets under my control.

Then the people got this superstition about how it mattered where your body was. I remember one time there were three people trying to fix the TV. By this time I would wait for some interesting thing they would do to fix the picture so I could trick them into thinking they had done it. One of the three guys had his hand in the middle of the TV screen. He was standing with one foot up on a chair. Seeing his hand accidentally rest in the middle of the TV screen, I took my cue and fixed the picture. One of the three guys announced, Hey, the picture's good. They relaxed. When the guy in front pulled his hand back, I made the picture go bad again.

The guy in back of the TV turning the dials on the back of the

TV said, "Let's all try to get our bodies where they were and maybe it will work again!"

A few seconds later, the guy in front rested his hand back on the middle of the screen and I did it again, fixed the picture. He tested it by pulling his hand away—I made the picture go bad—and then putting it back on the screen—and I made it go good again.

Then I noticed him take his foot off the chair and put it down on the floor. Again, I ruined the picture. When he put his foot back on the chair, he looked so startled when the picture went clear again. God, was I good to pull this off without ever getting caught.

He turned to the other students in the room and loudly announced, "Grounding effect." He had to have been an engineering student to have known a word like that back then.

The dozen or so students stayed for the second half hour of *Mission Impossible* with the guy's hand over the middle of the TV! And TVs were pretty small back then.

The only trouble is, I'd gone too far. For the next few weeks, virtually no one showed up in that TV room. They had had enough.

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Later in the year, they all came back again. So again I would play with this game, and just have so much fun. Sometimes people would have to pound the TV as hard as they could on top. Other times, there had to be three people on the TV at once—one pounding, one tuning, and one turning the color dial on the back that adjusted how much red, green, and blue the picture had. After that, they needed more than me to get the picture back! So a repairman had to be called.

After the repairman came, I heard someone at the TV mention that he'd said it was an antenna problem. I jammed the TV again, so what did they do? Of course, someone picked up the twin-lead

antenna wire and lifted it up over his head. I made the screen go good. He put it down and I made the screen go bad again. Up, good . . . down, bad. And after a while, I made it so he had to hold up the antenna higher and higher. This guy's trying to watch the last five minutes of some show, and he's stretched out to the ceiling, it was hilarious.

Except for Randy, I never told anyone else about it the whole year. I found it just amazing that at no time did anyone suspect that a human was toying with them. They never caught on! It was so funny. I couldn't make up a story this good. The only time I regretted using the TV Jammer in the TV room was during a daytime watching of the Kentucky Derby. Of course, I timed it down to the last stretch, and then I jammed the TV. Those kids erupted like animals, throwing chairs at the TV and everything. If it had been a human being, they would have beaten him to a pulp, they were that upset. And I felt horrible because I knew that if they had found me out on that day, it would've been hospital time.

There's a point where a joke crosses to a point where it is beyond funny—not funny anymore but scary—and this was it.

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I had a computer class at Colorado where I took the TV jamming concept a little further.

Just the fact that I was able to take a computer class was amazing. Back then, there were only a few colleges that had computer courses. Undergraduate computer classes were virtually unheard of, so this was a graduate class. Being enrolled in engineering at Colorado, even as a freshman, meant I could take any engineering class, even graduate classes, as long as I met the prerequisites. And luckily there weren't any for this course. This class was just amazing—in it, they taught everything about computers, their architecture, their programming languages, their operating systems, everything. It was such a thorough course.

The only problem was, it was held in the engineering building,

where the classrooms were really small. So only a third of our class got to see the professor in person in one room. The other two-thirds had to watch on TV, on closed circuit, in a room that had four TV sets on the wall.

So I thought, Okay. What a great opportunity for the TV Jammer. But first I had to make an even smaller TV Jammer, a version that would be even harder to detect. So I built one inside of a Magic Marker, including the battery and everything. (I'd taken the pen apart and put in a AA battery. At the very end of the pen I put in this little thumbscrew for tuning.)

I took it to computer class one day. I went to my usual seat over to the left rear of the class, and I took my little TV Jammer pen, turned it on, and tried to jam the TVs. I didn't know if I was going to be able to do this—I wasn't sure if it was even possible to jam TVs where the antennas came in on a coax. After all, coaxial cable was unusual in those days. The normal thing was to have twin-feed antennas.

But, sure enough, all the TV sets jammed. The one real near me didn't jam up that bad, but the other ones did. Well, almost instantly these three teaching assistants started looking at us. One of them said, "Okay. Whoever's got the transmitter, turn it off."

Wow. I didn't even know there were TAs in the class. So while they're looking right at us, saying "turn it off," do you think I'm going to reach my hand down and turn it off in plain sight? No way.

My plan had been to just jam it for a few seconds, but now I couldn't turn it off without getting caught.

So I'm sitting there kind of scared, afraid to move because they're watching us so closely. I couldn't even put my hand near it for fear that it might make the images on-screen wobble. I didn't even want to reach over to my Magic Marker and click the Jammer off because the guy next to me would hear me click something. He'd know I did it.

Eventually the TAs sat down, but they kept watching us. There was nothing they could do. And you know, the TVs weren't jammed so bad that we couldn't watch the professor or take notes. So our class just went on, with all of us watching the jammed TVs.

So I've got my Magic Marker TV Jammer sitting there between the two rings of my binder when suddenly the guy who's sitting the closest to the TV jammed the worst, in the right rear of the classroom, decides to gather his books and leave early. I decided to make the TVs waver as he was walking out. I felt like I could get away with it. I couldn't resist.

As he was leaving, the picture back there on the right rear TV went perfect. One of the TAs pointed at him. The TA said, "There he goes."

Pranks are entertainment, comedy. Not only did I manage to pull off this prank, but I managed to make it look as if someone else had done it. That's a step beyond the old rule "Don't get caught." I learned how to use that technique many times throughout my prank career. And if you're shocked that I can trick people with my pranks and not feel dishonest about it, remember that the basic form of entertainment is to make up stories. That's comedy.

I don't know if they ever did anything to that guy, but I doubt it. I hope not. It's not like they could catch him with a TV Jammer. As far as I knew, I had the only one.

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But I did end up getting in some trouble that year.

You see, I started writing programs that could kick paper out of the computer over at the computer printers everyone had to use at the Computer Center of the University of Colorado. That wasn't a big deal. But then I thought, Okay, what are computers for? They're for calculating numbers. Calculation has always been central to my association with computers, you know. So I tried to think up something really clever.

I wrote seven programs—they were all real simple but extremely interesting in a math sense. One of them dealt with what I called “magic computer numbers.” That would be the powers of two. So 2^1 equals 2, 2^2 is 4, 2^3 is 8, 2^4 is 16. These are the binary numbers all computers work with, so they are the most special of all the computer numbers.

I made it so the printer would print out the results formatted in a way that was readable. For instance, one line might say: 1, 2. That meant 2 to the first power is 2. The next would say 2, 4: 2 to the second power is 4. You will see that the numbers get really big really fast. For example, 2 to the eighth power is 256; 2 to the sixteenth power is 65,536. So pretty soon I am filling up pages with these really long numbers! After enough pages, the powers of 2 would be almost a line long. Then they would expand to two or three lines. Eventually it got to where each number might be a whole page or more!

Another program worked with Fibonacci numbers. These are numbers that go in a sequence like 1, 2, 3, 5, 8, 13, 21, 34 . . . Each Fibonacci number is the sum of the two numbers preceding it. So it's a never-ending sequence. All of my seven programs did this—calculated numbers in these long, ridiculously long, sequences.

Some programs have loops and don't stop running because there is a bug, or a problem, with a program. That is called an infinite loop, which I told you about in connection with the chess game I did back in high school. Anyway, the Computer Center automatically kicked off any program that ran more than 64 seconds. So I figured out that all my computers could print out 60 pages in under 64 seconds, and that's why I wrote each program to print out only 60 pages all numbered page 1, page 2, etc. The next time I ran the program, it would print the next 60 pages (beginning at 61), and so on. I wrote all my programs so they would punch some cards I could use the next time so the programs could pick up where they left off.

I would walk over to the Computer Center every morning and drop my seven programs off. Then, around noon, I would pick up my outputs and resubmit the programs. Then I would come back in the evening and resubmit them. I would get three runs a day times 60 pages times seven programs piling up in my dorm room. Mike, my roommate, started getting a little upset at all the space it was taking up. It was really piling up: reams and reams, feet and feet of computer paper, all stacking up in my dorm room.

Then, one afternoon, I got to the Computer Center for an afternoon run and they didn't have my programs there. There was a note there saying I should see my professor right away.

I went to see him in his office. He said, "Okay, sit down." He started a tape recorder—he punched a button and started recording us. I remember I got a bit scared.

"You've been running these programs on your own," he said.

And I said, "Yes. We were in a programming class. I was learning programming. I ran them under my own student number. I didn't try to hide the fact I was running them."

"This had nothing to do with our class," he said.

"It was FORTRAN," I told him.

"This is not the FORTRAN we teach," he said. And he was right. Because I had gone way into the manuals to find little

What Is FORTRAN?

FORTRAN is a computer language developed in the 1950s and still heavily used for scientific computing and numerical computation half a century later. The name comes from the words "Formula Translation." As a compiled language, it is typically faster and more powerful than an interpreted language such as BASIC.

tricks of mathematical symbols. I had gone way beyond simple programming, and we both knew it.

He said it took him a long time studying my programs to figure out what they did, but he finally figured them out. He said: "Are you trying to get me?"

Get him? I didn't know what he meant by that. I guess he felt threatened by the unrest happening in relation to the Vietnam War. The Students for a Democratic Society (SDS) was big on that campus. But I was totally apolitical except for registering once to be in the University Republicans Club! I mean, I was just a mild, meek engineering student and would never be involved in anything politically subversive.

"Out to get you?" I said. I had no idea what he was talking about.

Well, he picked up the phone and called someone at the Computer Center. "These programs . . . Mr. Wozniak should be billed for this computer time."

Then I found out what I'd done. I'd run my class five times over its annual budget for computer time. I didn't even realize there were budgets. I thought if you're in a computer class, you get computer time. That was logical to assume. But now I realized I ran up a whole lot of money on his account, and my best guess is he was using me to get out of it. I didn't think they would actually charge me, a student. A freshman. But I was scared because the amount he was talking about was in the thousands of dollars—many times the out-of-state tuition money.

So that's how it became very clear to me, at the end of that school year, that I was not going to make it an issue with my parents and try to go back to Colorado. I was on probation for computer abuse. I wouldn't let my parents find out. I didn't want them to get billed this huge amount of money. So that's how I decided to go to De Anza Community College the next year, instead of going back to Colorado like all my friends.

What really bothers me when I think of this now is, they shouldn't have charged me. They should've praised me for doing these brilliant programs all on my own.

And I did get an A-plus in that class.

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Now I was back home and attending De Anza Community College. I spent a lot of time designing and redesigning computers on paper, which is what I'd been doing in high school. Like when I took the manuals of popular minicomputers at the time (the pizza-box-sized, rack-mounted computers from Varian, Hewlett-Packard, Digital Equipment, and other companies in 1969 and 1970) and redesigned them, over and over, on paper so they would take fewer chips and run more efficiently.

By the time I finished at De Anza, I had literally designed and redesigned some of the best-known computers in the world. I'd become an expert on designing them, no question, because I'd redesigned their prototypes so many dozens of times. I'd done everything but build them. There was no doubt in my mind that if I ever did build them, I could get them to work. I was this virtual expert—and yes, I mean that in the software sense of the word "virtual." I never built those computers, but I was so entranced by and familiar with their innards that I easily could have taken any one of them apart and rebuilt it so that the computer would be cheaper, better, and more efficient.

I never had the courage to ask chip companies for free samples of what were then expensive chips. A year later I would meet Steve Jobs, who showed me how brave he was by scoring free chips just by calling sales reps. I could never do that. Our introverted and extroverted personalities (guess who's which) really helped us in those days. What one of us found difficult, the other often accomplished pretty handily. Examples of that teamwork are all over this story.

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Once, at De Anza, my quantum physics teacher said, "Wozniak. That's an unusual name. I knew a Wozniak once. There was a Wozniak who went to Caltech."

"My father," I said, "he went to Caltech."

"Well, this one was a great football player."

That was my father, I told him. He was the team's quarterback.

"Yes," the teacher said. "We would never go to football games, but at Caltech, you had to go just to watch Jerry Wozniak. He was famous."

You know, I think my dad was the one good quarterback Caltech ever had. He even got scouted by the Los Angeles Rams, though I don't think he was good enough to play pro. Still, it was neat to hear from a physics teacher that he remembered my dad for his football. It made me feel like I shared a history with him. The teacher once even brought me a Caltech paper from back in those days with a picture of my dad in his uniform.

I didn't get along with all the teachers, though. I was taking an advanced-level math class, and the teacher caught me not paying attention. (I was trying to figure out how to write a FORTRAN compiler in machine language for the Data General Nova.)

I was just at the first line where you have to enter something and store it in memory when he said: "You've got so much potential, Wozniak. If only you'd just put yourself into this material."

It stung me the way he said that in front of the whole class. That wasn't necessary. I just wanted to sit in class and do whatever it was I felt like doing. Maybe I was bored, I don't know. I was the sort of person who read the book, took the test, and got good grades in subjects like math.

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It was also at De Anza that I got this mental turnaround on politics. I started seriously thinking about whether the Vietnam War was right or wrong. Who was it helping, and did we have any place there?

Back in high school, I was for the war. My father told me our country was the greatest in the world, and my thinking was like his: that we had to stand up for democracy versus communism; and the reasons why, stemming from our Constitution. I had never thought deeply about political issues aside from that, and I was really for my country, right or wrong. I mean, I was for my country the same way you root for your school's team, right or wrong. At the University of Colorado, the University Republicans Club was one of the only two clubs I joined (the other was the Amateur Radio Club).

But I started to wonder why so many people were protesting the war so visibly. A lot of academics and journalists were talking about the history of the Vietnamese people and had explanations for why the U.S. position was wrong. It was a civil war, involving treaties, agreements, and a history that didn't affect the United States one whit. The trouble is, I could find no intelligent academic reasoning coming from the pro-war side, just the constant refrain that we were doing good. They could only say that we were there protecting democracy.

One of my biggest problems was that South Vietnam, which we were supposedly protecting, wasn't even close to a democracy. It was more like a corrupt dictatorship. How could we ever stand up for a dictatorship? I started seeing that there was a lot more truth on the side of the people against the war.

The people against the war were also talking about how good peace was compared to war. Sure, the world can't live in perfect peace and harmony, but it's a good ideal. I had come to learn of Jesus, from my friend Randy Adair in college, that he always tried to find ways toward peace. Although I'm not a Christian per se, and don't belong to any religion, what Jesus the historical figure stood for were things I stood for, and those stories Randy told me about him struck a chord with me emotionally. I didn't believe in violence or hurting people.

At De Anza, I thought deeply about the war. I considered myself to be athletic and brave. But would I shoot a bullet at another human being? I remember sitting alone at the white Formica table in my bedroom, coming to the conclusion that I could let someone shoot at me, but I couldn't shoot back.

I thought, What if I'm in Vietnam and I'm shooting at some guy? He's just like me, that guy. He sits down just like I do. He plays cards and he eats pizza, or the equivalent of it, just like normal people I know. He has a family. Why would I want to hurt this person? He might have his reasons for being where he is in the world—and Vietnam had its reason—but none of these reasons ever touched me in California.

From that standpoint, I could see how this war could be a pretty dangerous one for me. Because I was morally and truthfully a conscientious objector in every sense. But the military only counted you as one if you were in a church (which made you exempt from conflict duties), and I had no church. I had no religion. I just had my own logic.

So I wasn't a conscientious objector, I just objected to my personally having to kill or hurt anyone.

Cream Soda Days

When I was about nineteen, I read the Pentagon Papers and learned what was really going on in Vietnam. As a result, I started to have some major conflicting feelings and some nasty fights with my dad.

By then he was drinking heavily, and he wasn't the greatest opponent to argue with. But I had a new truth that replaced the old one even more strongly. I started to believe in peace. And I began to realize how far governments would go in order to get people to believe them.

For one thing, the Pentagon Papers showed what the CIA and the Pentagon people truly knew, and that the president was being carefully coached to put words together and lie to the American people. He was saying the opposite of the truth to trick the American people into thinking they should actually support the war. For instance, the papers got right down to the Gulf of Tonkin incident—which never happened the way the government said it did. The papers also said how, in every battle, the public was always told that ten times as many Viet Cong as Americans died, despite the fact that we had no way to count them. And most Americans believed this crap. The Pentagon Papers documented this deliberate deception.

Learning about that was one of the hardest things I had to deal with in my life. You see, I just wasn't raised to believe that a democracy like ours would spread these kinds of lies. Why was the United States government treating the American people as the enemy and purposely duping them? It made no sense to me.

And the worst thing that came after that, for me, wasn't the Vietnam War itself, but the pain and stress it caused people. That's because, as I was becoming an adult, I started gaining a new ethic—a profound care for the happiness and welfare of people. I was just starting to figure out that the secret to life—and this is still true for me—is to find a way to be happy and satisfied with your life and also to make other people happy and satisfied with their lives.

Even in high school, where I believed in truth with a capital T, I was willing to change my beliefs if someone came along to show

The Gulf of Tonkin

Not everyone reading this is going to remember this incident, but finding out about it was instrumental in changing my own feelings about the Vietnam War.

The Gulf of Tonkin incident was an alleged attack on two American destroyers (the USS *Maddox* and the USS *C. Turner Joy*) in August of 1964 in the Gulf of Tonkin by North Vietnamese gunboats. Later research indicates that most of the attacks did not actually occur.

According to the Pentagon Papers and various reports, the attacks were pretty much made up by President Lyndon B. Johnson's administration. The U.S.-supported South Vietnamese regime had been attacking oil-processing facilities in North Vietnam, but it was the CIA that helped plan and support it in order to give the administration a good reason to involve the U.S. in the conflict.

me something better. That's what the Pentagon Papers did for me. They pointed out that even the president was subject to the pressures of the military-industrial complex, the major institution of our land. And after reading this, I decided not to vote, that it wouldn't matter either way. I figured that pretty much I'd get the same life no matter who was elected. I thought it was better not even to go into the voting booth.

But I did vote a couple of times. I voted for a guy named George McGovern, who promised he'd find a way to stop the war. I voted for Jimmy Carter, because the words he spoke seemed to come from the same philosophical point of view as my own. He believed, as I did, that war was a last resort and not a first.

I voted for George W. Bush in 2000, because I thought it would be nice to have an average Joe kind of person in the White House instead of a smart, well-educated one. Someone who could only speak in very small words. Okay, I'm joking. The fact is I voted for Ralph Nader. But since all the pundits said that a vote for Nader was a vote for Bush, I now tell people I voted for Bush just to watch their jaws drop.

Seriously, though, I still think about this whole era with a lot of pain. Being brought up by my dad, who'd taught me that we had the best government in the world and that our government was the best one there ever could be even with its flaws, well, that kind of fell apart. He told me the purpose of the government was to take care of its people and make things better for them.

During the Vietnam War, of course, there was a mandatory draft. When you turned eighteen, you had to register. If you were a college student, you would get what was called a 2S deferment; otherwise you would be classified as 1A. That meant that any day the military could draft you and send you off to boot camp.

Once you were 1A, the government had a year to draft you. After that, you would be exempt. That's why not everyone who was 1A wound up serving.

I submitted a report card to the San Jose draft board in order to get my 2S deferment, but I didn't submit one of the proper forms the government required to show you were a college student. By mistake I only sent in my report card.

A couple of months later, a big delay, I received notice that the San Jose draft board had voted five to three to make me 1A. What? But I was a student.

This is when I decided that I would go to prison or Canada or, more likely, try to get let off by a judge rather than go to Vietnam. In fact, a judge in San Jose—his name was Judge Peckham—had let a couple of guys off as conscientious objectors despite their lack of membership in a church.

One of those people let off had been one of the brightest math stars at my high school, Allen Stein. Quite a coincidence. So I had good reason to expect the same in my situation.

So since I was 1A anyway, I took a year off from school to program computers and earn money to pay for my third year of college and buy a car.

Then an amazing thing happened. The U.S. Congress created a draft lottery. That meant that those of us who were 1A would know the likelihood of our being called up. This was so it wouldn't be random. That way, you would know your chances—and I thought it was great. It helped me plan my life.

The way the draft lottery worked was your birthday determined what order you got called in. They would assign everyone's birth date to a number from 1 to 366. So January 1 might be 66, January 2 might be 12, it was totally random.

Well, during the week before they announced the results of the lottery, I got a feeling I have never had before or since. A feeling of physical warmth, like I was going to be protected and get a high number in this lottery. I had a stronger level of certainty than I would ever have let myself feel about the unknown. I can't explain it. I'm not a superstitious person in any way. I have

always believed in reality, the truth, and the provable. But this was so certain in my head. I rode my bike around, just smiling and smiling and smiling about it. I couldn't stop. It was a wonderful, positive feeling, and I couldn't ignore it and pretend it wasn't there.

And sure enough, I read in the newspaper the day of the draft lottery that I got number 325. A great number! That meant it was virtually certain I wouldn't be drafted. It's so weird. I got such a great number, but I wasn't even surprised or elated. I felt like I'd known it all along. The feeling I'd had was that strong.

But then something terrible and unexpected happened.

About a week after I got my draft lottery number, I got a letter from the San Jose draft board. It said—in one sentence—that they were granting me a student deferment after all.

This, after taking months to notify me that they'd voted five to three not to grant my student deferment when I deserved it, was bad enough. But worse, it also meant that in a later year they could make me 1A a second time.

I stood there with the letter in my hand, stunned. They were playing tricks with my life. Dirty tricks. They used the application I'd made for a student deferment as an excuse to grant it to me now, knowing that I already had a great 1A number.

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From that point on, I saw that the government would do whatever it could to beat a citizen, that it was just a game. And this was the exact opposite of the way I had thought of government my whole life. That episode taught me an important lesson about government, authority, even the police. You couldn't trust them to do the right thing.

Now I had to go back to the draft board and request to keep my 1A—which was what I'd had anyway—and keep the same number. Luckily, they agreed.

I can't even describe to you the shock and disgust I felt at our

government: that they would play this kind of game with my life, that they didn't care about people the way my dad had taught me. I'd thought the government was here to protect us, but that turned out to be wrong. I now believed the government was just out to do what was good for the government and would lie about anything they could get away with. They were not there to do sensible things, and they played with my life in the worst possible way.

From then on, my dad and I were at complete odds. I never trusted authority after that. That's too bad, because since founding Apple and all, I've met lots of good people in the government. But still, this hangs over me. I can hardly trust anything I read.

So between the time I was a kid, when my dad taught me extreme ethics, and the time I realized what was going on with the Vietnam War, I changed profoundly, a full 180 degrees. I became skeptical. I stopped believing blindly in things. It was a major turning point. I lost the trust I'd always had in institutions of all kinds and it has never really returned.

I swore to myself I would put up my own life before letting something like the Vietnam War ever happen again to young kids.

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Maybe you've seen pictures of me from the early days and thought I looked like a hippie. I guess I did, a little. But let me tell you, I was never a hippie.

I tried to be a hippie, but I could never be what they were—not in high school or even in college when all that protest stuff was going on. I'd try to hang out with hippies because I stood with them politically, but they'd usually ask me to leave because I wouldn't use drugs. I still wanted to hang around them because I felt my mind was so open—as open as theirs were—and I got what they were saying. I wanted them to be open with me, but their drugs got in the way. They didn't trust me because I wouldn't do drugs with them.

But I believed in almost everything they were trying to do. Everything I was reading about hippies and hippie beliefs in the late 1960s—the free love movement, things like putting flowers in guns—I knew that was me and what I wanted to be. I agreed with every bit of it. I believed, like hippies did, that everybody should be able to get along and help each other out and live out whatever kind of existence they wanted. And I believed it could be an existence without structure and without laws and without organization and without politics.

People should just agree to live together and be good people, I really believed that. I was tremendously influenced by these kinds of hippie thoughts, these kinds of philosophies.

I would wear this little Indian headband, and I wore my hair really long and grew a beard. From the neck up, I looked like Jesus Christ. But from the neck down, I still wore the clothes of a regular kid, a kid engineer. Pants. Collared shirt. I never did have the weird hippie clothes. I was still middle ground; I was still the way I'd grown up. No matter how hard I tried, it was like I couldn't get outside of normal. Hippie is a way of life, not just a matter of clothes and hair, and I didn't lead that kind of life. I didn't live in weird little places with no money with weird curtains hanging in my windows. And I didn't do drugs. I wouldn't.

At the time, not doing drugs or drinking made me real different. I mean, at the time, especially during my second year at De Anza and for years and years after, people would say things like: "Oh, using LSD can really expand your mind." I remember a guy—John was his name—who claimed that all the A's he got were when he was on acid.

But I thought to myself: Well, if drugs are really better for your mind and can make you think better, then wait a minute. When you take a drug, it's you plus the drug that's working, right? It's not just you. And I really, really wanted to be successful in my life just based on me and my mind alone. I knew that I was bright and

that my brain was going to take me places. I didn't ever want it to be an equation that amounted to a result coming from my brain plus something else. I wanted to be judged on my own abilities, on what I did and what I thought, and that alone. So that was pretty much my view on drugs, and I never did any of them.

As for drinking, I didn't even get drunk until I was thirty, in 1980. It was on my first flight out of the United States to Sri Lanka. I was extremely scared on the plane so I was drinking. I wasn't sure they'd let you off a plane if you were drunk. I managed to walk off the plane without assistance, and I ended up telling a really awful joke to a customs official:

A lady who'd never seen an elephant before saw an escaped one in her garden. Shrieking, she called the police. "There's a huge animal in my garden!" she said. "Pulling up the vegetables with his tail! And you wouldn't believe where he's putting them!"

I don't remember if he laughed or not. I don't think so. It's not a joke I normally would have told. It's sort of dumb and hard to get.

Anyway, I never liked alcohol. It made people act noisy and out of control. My dad, for instance, he used to drink martinis. I always noticed how he reasoned differently when he was drunk. Especially as he and I got older, I thought it got really out of hand, the way he would get drunk and yell at my mother. That's not the way he was when he wasn't drinking.

So I didn't drink or do drugs, and as I said, that usually made it kind of weird for the hippie people I wanted to be friends with, people who thought like me in every other way. What a sad thing. During that second year at De Anza, I remember driving my first car down to Santa Cruz. This was back when there were hitchhikers all over the place. (The car I was driving was this purple convertible I'd named Hubbs after a weird chemistry professor I had, but it wasn't that funny a joke because neither the car nor the professor was really that weird.)

Anyway, I stopped and picked up a group of people. They were

definitely hippies. And I took them down to Santa Cruz. We're hanging out on the boardwalk and I notice that one of them, a young girl sitting on a bench, was breast-feeding. Breast-feeding! I'd never seen anything like that before in my life! I just turned my head away really quickly, but it made such an impression on me. I started talking to her, and immediately fell in love with her and her baby. It turned out that she and her baby and a bunch of people all lived together in this commune near me in Sunnyvale. Later I would ride my bike out there a lot and stop at a park near their house and read books. I would go over and hang out with them. We'd eat and do the ohm chanting and all of that. And they would take me with them to meet all these Eastern philosophy-type teachers, really getting me exposed to Eastern thoughts of peace and quietness. I listened to these principles of meditation, and I would just sit down and try to get my head into a quiet place by myself.

The sad thing was, eventually even these hippies didn't want to hang around me anymore. It made them uncomfortable that I didn't do drugs.

So this was a hard social time for me. I remember that at one point I was taking some night classes at San Jose State and this pretty girl comes up to my table in the cafeteria and says, "Oh, hi." She just starts talking to me, and I'm so nervous all I can think to ask her is what her major is. She says, "Scientology." I'd never heard of this, but she assured me it was actually a major and I believed it.

She invited me to a Scientology meeting, and of course I went. I ended up in the audience watching this guy make this incredible presentation about how you can basically be in better control of yourself and that you could get really happy from that.

After the meeting, the girl I met sat with me in some little office for an hour, trying to sell me these courses to become a better person. I was going to have to pay money for them.

I said to her, "I've already got my happiness. I've got my keys to happiness. I don't need anything. I'm not looking for any of this stuff." And I meant it. The only thing I might've wanted was a girlfriend, that's for sure, but the rest of the stuff I already had. I had a sense of humor, and I had this attitude about life that let me choose to be happy. I knew that whether to be happy was always going to be my choice, and only my choice.

Plus I had these values, values I'd grown up with. I already had this peaceful feeling inside my head. To this day, I'm one of those people whose head just floats. I really did feel happy most of the time. I still do.

So the bottom line, of course, is she never sold me any courses or anything. In fact, she just walked out and never came back. When I wasn't interested in buying her courses, she left and just let me sit there. I sat and sat, waiting for her to come back. Finally I walked out the door and left, too. I thought, Man, that's too bad. She was just about sales, that's all she was.

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After my year at De Anza, I decided the thing to do was to take a job where I could actually program computers. I thought I'd skip a year of school so I could earn enough money to go for a third one, at Berkeley maybe.

Now, for a while I'd been telling my dad that I definitely was going to own a 4K-byte Data General Nova someday. That was just enough kilobytes needed to program in. It was this huge, powerful computer at the time. I loved its internal architecture and everything about it. I even had a poster of it up in my room. So I heard there was a place in Sunnyvale that sold these Data General computers. My friend Allen Baum and I drove down to the place.

Well, the office was beautiful, and right in the middle of the lobby was this big glass display with a large computer in it. It wasn't a room-sized computer like a mainframe, but a midsized

computer. It was about as large as a refrigerator, with other things like large printers and disk drives the size of dishwashers attached to it. There were some wires hanging loose with engineers working on them. And I thought, Wow, here's a computer actually being designed and built. That was a shock for me to see.

Another shock turned out to be that I had walked in the wrong door, that I wasn't at the Data General company at all, but at a smaller company called Tenet. Allen and I both filled out applications for jobs as programmers—and you know what? We got them.

We got to program in the language FORTRAN, and also in machine language, which is nearest to the lowest-level language (1s and 0s) a computer can understand. We got to know that computer so deeply that summer. We really got into the depths of its architecture. Personally, I didn't think much of the architecture inside, although they ended up building something pretty good—a working computer, a fast computer, a low-cost computer for what it was. I mean, it cost more than \$100,000, and those were 1970 dollars. I was impressed by that. It had an operating system that worked well and several programming languages.

Now, of course, in no way was that Tenet computer like our computers today. It had no screen for a display and no keyboard to type into. It had lights you had to read off a front panel, and it took information from punch cards. But for the time being, yeah, I guess it was pretty cool.

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Tenet actually went out of business the next summer—I stayed for the duration, having decided not to return to school that year after all—but my time there turned out to be really fortunate.

You see, during the summer, I remember telling one Tenet executive how I had spent the last few years designing and redesigning existing computers on paper but could never build one because I didn't have the parts.

One time, at my old friend Bill Werner's house, I got Bill to call up a chip company, but he could never get them to give us free parts, never. But I asked this Tenet executive, and he said, "Sure, I can get you the parts." I guess he had access to sample parts, and that was what I needed.

To help him avoid having to get me tons and tons of parts—parts I would need to build some kind of existing minicomputer—I decided I would build a computer that was just a little one with very few chips.

I'm talking about, like, about twenty chips—which is very, very few chips compared to the hundreds it would have taken to build a normal computer at the time.

Now, I had this other friend, Bill Fernandez, who lived down the block. I started hanging around at his house, and we just started putting together this little computer I designed (first on paper, of course) piece by piece, bit by bit. He helped me by doing all kinds of things—like soldering, for instance.

Anyway, we would do this in his garage, and then we'd ride our bikes down to the Sunnyvale Safeway, where we would buy Cragmont cream soda, and then drink it while we worked on this machine. That's how we started referring to it as the Cream Soda Computer. All the Cream Soda Computer was, really, was a little circuit board that allowed you to plug in connectors and solder the chips I had to the connectors. This board was tiny—I would say it was no larger than four to six inches.

Like all the computers at the time, there wasn't a screen or a keyboard. No one had thought of that yet. Instead you'd write a program, punch it into a punch card, slide it in, and then you'd get your answer by reading the flashing lights on the front panel. Or, for instance, you could write a program that would tell the computer to beep every three seconds. And if it did, then you would know it was working. It turned out just as I had designed it, with few chips because I didn't want to ask that executive for

too many free samples. So it was just the most minimum thing you could even call a computer. What I mean by that is, it could run a program. It could give you results.

The other significant thing about it was the fact that it had 256 bytes of RAM. (That's about the size in memory a word processor would take today to store this very sentence.)

RAM chips were almost unheard of at this time. Back then, almost all computers had a type of memory called "magnetic core memories." When you dealt with them, you had to deal with messy voltages to spike the right currents down the wires, the wires that had to go into these little round magnetic cores that looked like tiny donuts you needed a magnifying glass even to see. This was definitely not the type of electronics I had in mind. With RAM chips, though, you just plug them in and connect them to the CPU, the brain of the computer. You connect them to the processor with wires and that's that. So, as you can see, I was extremely lucky to get those eight chips that added up to 256 bytes. And as I said, even then you couldn't do much of anything in that small a space.

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What Is RAM?

RAM, short for random-access memory, was a new type of computer storage back in 1970. These are chips whose contents can be accessed in any (i.e., random) order. All computers today have RAM chips inside to store data—not permanently, but while your computer is on and you're working. When the computer shuts down, the contents of RAM goes away. That's why you need to save your programs to disk.

One day my mom called the *Peninsula Times* newspaper and told them about the Cream Soda Computer. A reporter came over and asked some questions about it and took some pictures. But just as he was finishing, he accidentally stepped on the power supply cable and blew out the computer. The Cream Soda actually smoked! But the article ran anyway, and that was pretty cool.

But you know what? I knew deep inside that it didn't matter that I had built this computer. It didn't matter because the computer couldn't do anything useful. It couldn't play games, it couldn't solve math problems. It had way too little memory. The only important thing was that finally, finally, I'd been able to actually build a computer. My very first one. It was an extraordinary milestone in that sense.

Five years later, companies would be building and selling computer kits that were just about at this level—they had the same amount of memory and the same awkward front panel of lights and switches.

Looking back, I see the Cream Soda Computer as kind of a jumping-off point for me. And I got there early.

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One other thing: the Cream Soda Computer turned out to be the way I first met Steve Jobs. I was four years ahead of him in school so I didn't know him; he was closer to Bill Fernandez's age. But one day Bill told me, "Hey, there's someone you should meet. His name is Steve. He likes to do pranks like you do, and he's also into building electronics like you are."

So one day—it was daytime, I remember—Bill called Steve and had him come over to his house. I remember Steve and I just sat on the sidewalk in front of Bill's house for the longest time, just sharing stories—mostly about pranks we'd pulled, and also what kind of electronic designs we'd done. It felt like we had so much in common. Typically, it was really hard for me to explain

to people the kind of design stuff I worked on, but Steve got it right away. And I liked him. He was kind of skinny and wiry and full of energy.

So Steve came into the garage and saw the computer (this was before it blew up) and listened to our description of it. I could tell he was impressed. I mean, we'd actually built a computer from scratch and proved that it was possible—or going to be possible—for people to have computers in a really small space.

Steve and I got close right away, even though he was still in high school, remember, and lived about a mile away in Los Altos. I lived in Sunnyvale. Bill was right—we two Steves did have a lot in common. We talked electronics, we talked about music we liked, and we traded stories about pranks we'd pulled. We even pulled a few together.

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When I met Steve Jobs I was still hanging out with this other guy I'd known since high school, Allen Baum.

Allen was kind of a nerdy, skinny guy with glasses when I first met him back in high school. We were both in the super-elite of students, not just the ones in top classes but students who outperformed pretty much everybody else. We'd be selected out by teachers to compete in math contests or go to speeches and lectures, that kind of thing. So we all knew each other. Most of us were considered by other kids to be kind of weird outsiders, and Allen was even smaller, scrawnier, and more outside than I was. He was even nerdier.

Later he came to be very into hippie things and San Francisco-type music like the Grateful Dead and Jefferson Airplane, but he started out just completely way, way on the outside.

From high school on, I used to like to go visit Allen and his parents. They were Jewish with relatives who'd died in concentration camps; it was shocking and all so new to me. Allen's father, Elmer, was an engineer who loved humor—he was incredibly

funny—and he was really active in civil rights causes. His mom, Charlotte, was like that, too. I thought of Elmer and Charlotte Baum as being so much like me—just kind of casual and fun.

So, as I said, I was hanging out with Allen a lot when Steve Jobs, who by now was a junior at Homestead High, had an idea. He wanted to create a huge sign on a giant bedsheet with a flip-off sign—you know, the middle-finger salute—right during graduation. He thought the sign should say “Best Wishes.” We started calling it the “Brazilian Best Wishes” sign.

So we went right to work. We got this big sheet—it had been tie-dyed because Allen and his brothers were always tie-dying everything back then—and spread it out in Allen’s backyard. Anyway, we started sketching out our drawing with chalk—a big hand with its finger sticking up. And Allen’s mother even helped us draw it—she showed us how to shade it so it looked more like a real hand, less like a cartoon. I remember how she sort of realized what the hand was doing partway through, but she just snickered at us and smiled, saying, “I know what that is.” But she didn’t stop us. I guess she didn’t know what we planned to do with it, exactly.

On the sheet, we signed it “SWAB JOB. The S and W stood for Steve Wozniak, the A and B stood for Allen Baum, and JOB stood for Steve Jobs. We finished the sheet and rolled it up. Late that night, we climbed onto the top of the C building, where we planned to drop it. The idea was that we would attach it to this forty-pound fishing line and kind of pull it down when the graduating Homestead High seniors walked past.

Well, we practiced it and found out that you just can’t pull a sheet down off a roof and have it roll down nicely. It doesn’t come down off the roof easily, and it will pull other junk off the roof, and it can come down all weird in different shapes.

So the next night we tried to make this little cart thing with an axle and two wheels that we could pull. The idea was that it

would let the sheet down gradually. The axle was nearly eight inches wide. But we found that one of the wheels would always get stuck on its little track. We just couldn't get it right.

By the fourth night of trying to do this, Allen and I were working alone. Steve just didn't have the stamina to stay up and work all night. And by then we'd come up with another idea. Don't use the axle, but keep the wheels. We rigged up this little hookup on the building, higher than the sheet would be, and attached the fishing line and a couple of skates to it. We then tested it out. We stood on the roof, let go of the fishing line, and watched the little skates roll down their ramps until they pulled the sheet down so it scrolled down, pulling down the left side and the right side together. Left to right. It worked perfectly.

We almost got caught that night, by the way. We tried to test it again, but a janitor came along. We just ducked on that roof and were lying down as low as we could. I remember the janitor moving his flashlight around and the light landing on my hand. But before he could call anyone, we ran out of there like crazy.

A couple of days later was graduation day. I woke up that morning to the phone ringing. It was Steve calling from school with bad news. It turned out someone, probably another student, had cut the fishing line and pulled the sign down that morning. So Steve got in trouble—I guess the "SWAB JOB" gave it away, and we never got to play our prank.

Afterward I thought about this a lot. I finally came to the conclusion that even though our "Brazilian Best Wishes" sign didn't come off, it wasn't a failure. Some projects are worth the energy and worth spending a lot of time on, even if they don't come out perfectly.

I learned about teamwork and patience and hard work from that prank—and I learned never to brag about my pranks. Because I found out a year later that Steve Jobs had shown some of the students our prank, showing off. And the guy who told me

that—that Steve Jobs had shown him the sign—said he was the guy who cut it down.

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Steve and I were into listening to Bob Dylan and his lyrics, trying to figure out who was better, Dylan or the Beatles. We both favored Dylan because the songs were about life and living and values in life and what was really important. The Beatles mostly made these nice little happy songs—you know, nice-to-know-you, nice-to-be-with-you, nice-to-be-in-love-with-you songs. They were simple—even after albums like *Rubber Soul* came out. The songs the Beatles did were not as deep down and affecting your soul and emotions as Dylan's were. They were more like pop songs. To us, Dylan's songs struck a moral chord. They kind of made you think about what was right and wrong in the world, and how you're going to live and be.

At any rate, this first introduction we never forgot, and later on Steve and I were really linked. Linked forever.

Chapter 6

Phreaking for Real

In 1971, the day before I headed off to my third year of college at Berkeley, I was sitting at my mother's kitchen table and there happened to be a copy of *Esquire* sitting there. Even though I never usually read this magazine, for some reason I started flipping through it that day.

I came upon an article called "Secrets of the Little Blue Box." Those were interesting enough words to make me stop skimming and read that article all the way through.

Now, it was labeled as an "incredible story", and I had no idea what a Blue Box was until I started reading the article. But as soon as I did, it just grabbed me. Wow! You know how some articles just grab you from the first paragraph? Well, it was one of those articles, probably because this story was about tech people like me. Back then, there were never articles about tech people—really, never—so once I started reading this article, about people like me, I couldn't stop. It basically was a story about how a bunch of technical kids and young engineers around the country had figured out how to crack codes on the phone system. The article called them "phone phreaks." These people were able to figure out that by just whistling certain tones into a phone handset, they could make telephone calls within the Bell phone system for free.

Essentially, they first would dial an 800 or 555 number, any free exchange, then they would make this tone sound to seize the line. If this certain tone worked, they'd get a chirp that meant they were now in control of a piece of phone circuit equipment called a tandem. (A tandem just waits for special tones to direct calls throughout the phone system.) The phone phreak could then give the system the tones it needed to dial any seven- or ten-digit number just by making a bunch of certain sounds that were equal to the numbers on a phone from Ma Bell's perspective.

That sounded plausible in a way. I already had a basic idea about how the tone system worked on telephones. And the people in the article—in this “incredible story”—were claiming that by doing this, they were finding out things about the phone system no one knew about. I'm talking about things like its inherent bugs and holes and weaknesses and, of course, all the ways to take advantage of them. So, as I said, they were doing things by whistling tones into the phone lines, by tricking operators, by bouncing calls off satellites and back to other countries. They were doing all this stuff. And though it seemed an unbelievable story, I kept reading it over and over, and the more I read it, the more possible and real it sounded.

The other thing that intrigued me about the article was the fact that it described a whole web of people who were doing this: the phone phreaks. They were anonymous technical people who went by fake names and lived all over the place. Some were in the Northeast, some in the Southeast, some were in the West. Just all over. The story told a tale about some guys who drove out to Arizona, clamped a wire to a pay phone, and were somehow able to literally take over the whole country's phone networks. It said they were able to set themselves up into ten-way conference calls.

The characters sounded just too perfectly described not to be true. I remember how it spoke about some blind kids who just

wanted someone to talk to. Somehow they'd gotten the phone company people to tell them some of the secrets of the phone company and were using them to talk to each other. That made sense to me, too.

The article also talked about the ethics these guys supposedly had. That it wasn't just about free calls. One of the guys said he was basically trying to do a good thing by finding flaws in the system and letting the phone company know what they were. That appealed to me.

The article also talked about one of the secrets these guys had discovered. Well, I already knew this secret, so I guess it was kind of a rediscovery. I'm talking about the technique of taking any phone—you can do it with any phone to this day—and tapping out phone numbers with the hook switch. What I mean is the actual switch on the phone that tells the phone company if the phone is on or off the hook. What you do is pick up the phone. You hear a dial tone, right? Then if you click that hook switch once, it's like dialing a "1." Click it twice really quickly and that is the same as dialing a "2." Clicking it ten times in a row would be the same as dialing a "0." (The reason this works goes back to the old days of rotary phones when you dialed a "5," and the dial would swing back five times—click click click click click.) And like I said, the system still works like that to this day. Try it.

But this was a trick only a very few people knew back then. So I could tell that the characters being described were really tech people, much like me, people who liked to design things just to see what was possible, and for no other reason, really. And because I knew the hook switch thing, too, I immediately got intrigued.

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In the *Esquire* article, there was a phone phreak named Joe. He was blind. According to the article, he'd discovered this cool thing: that if you play a really high E—two octaves above the

high E on the guitar, for example, which is 2,600 hertz (Hz) exactly—it was the exact tone that seized the tandem and gave you control of the phone system. It probably still works to this day, and you can try that, too. Anyway, Joe was able to actually make this whistle sound with his mouth!

Now, Joe had perfect pitch—probably because he was blind, I don't know. His first whistle seized the line, and then he could make a bunch of short whistles to dial numbers. I couldn't believe this was possible, but there it was and, wow, it just made my imagination run wild. Because just by whistling this high E, he could from there dial a long-distance call that would then be free. To the phone company, it would look like a free 800 or 555 long-distance phone call. And he was doing it all with his mouth!

The *Esquire* article also described someone who went by the name of Captain Crunch, after the cereal (Cap'n Crunch), which used to have a whistle toy in it. Captain Crunch used the whistle and discovered the same thing the blind phone phreak did: that if you plugged the right hole in the whistle and hit that high E, that 2,600 Hz sound—it blew just the right note that basically seized the phone line for anything you wanted to do.

To make a call after seizing the line, Captain Crunch used a device the article called a “Blue Box.” It put pairs of tones into the phone, similar to the way touch-tone phones work. This method worked everywhere on the multifrequency (MF) system in the United States, where Joe's and the cereal box's whistle only worked in a few places with old single-frequency equipment.

In the story, the guy who built the Blue Box supposedly stole or had loaned to him a standard phone company manual that listed all the frequencies he would need to build it. The article said the phone company figured it out and started to withdraw all those manuals from every library in the country. They made it secret, in other words. They wouldn't let it out anymore. But you

know what? The secret was already out. Way out. Too late for the phone company, or so the article said.

This idea of the Blue Box just amazed me. With it, you could just hook into some 800 number and use that line to make one call after another, all over the world. It didn't plug into the handset or anything. It was very simple. You just put its speaker up to the mouthpiece of the phone. Although it was incredibly easy, only something like a thousand people in the country, technical people like me, could ever have figured it out and used it.

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One of the first things I did after reading the article was to call up my friend Steve Jobs. He was just about to start twelfth grade at Homestead High School, the same high school I'd gone to. I started telling him about this amazing article, and how it made sense, really made sense, in a technical way. I told him that, according to the story, the whole system was grabbable. Attackable. And I told him how these smart engineers portrayed in the story overtook and used it. They apparently knew more about the phone system than even the phone company's engineers. If the article was right, and I thought it could be, that meant all the secrets of the phone company were out. It meant people like us were starting to create little networks in order to exploit them.

This was just the most exciting thing for us two young guys to be talking about. I was twenty; Steve was probably about seventeen at the time.

And while I was on the phone with Steve that afternoon, I remember I just stopped mid-sentence and remarked, "Wait, Steve, this article is just too true. They put in real frequencies like 700 Hz and 900 Hz. They even gave the way to dial a '1,' a '2,' and a '3.' And they even gave the codes for dialing a call all the way to England."

Steve and I came up with a plan to give it a shot.



An hour later I picked Steve up and we drove down to SLAC, which is pronounced “slack” and is short for the Stanford Linear Accelerator Center. It had a great technical library, just tremendous. It had the kinds of technical books and computer books and magazines you wouldn’t encounter in normal libraries or any other place I knew. If there was any place that had a phone manual that listed tone frequencies—the manual the phone company was trying to pull out of circulation—this would be it.

Anyway, I snuck into this library on many Sundays during my high school and early college computer design days. I never felt like it was sneaking, exactly, because they always left the doors open. In my experience, I’ve found that smart people often leave doors open. Maybe it’s because they have other things on their minds.

So Steve Jobs and I snuck in that day, that Sunday in 1971, and started searching for books with telephone information. Like I said, that *Esquire* article gave tons of details, not only on how tones are used to dial numbers, but also on how the tones were in pairs. For instance, it said that the 700 Hz and 900 Hz tones together stood for a “1,” that 700 Hz and 1,100 Hz meant a “2,” that 700 Hz and 1,300 Hz together meant “3,” and so on. There was even more detail than this, details I figured we could check right there at the SLAC library. So Steve and I sat in there looking for confirmation that this Blue Box thing was for real—we wanted the complete list of tones that could theoretically make all the digits. Because that would mean we could build one.

We were individually flipping through various books and I had a blue one in my hand, maybe two inches thick, with some phone system reference like the CCITT Handbook. In case you’re wondering, CCITT is a long-forgotten acronym that stands for the Comité Consultatif International de Télé-

graphique et Téléphonique. It's the French name for an international standards-setting group for telegraph and, later, telephone systems.

I was flipping around and suddenly stopped on a page. There it was: a complete frequency list for MF (multifrequency) telephone switching equipment.

Sure enough, just as the *Esquire* article said, a "1" was composed of 700 Hz and 900 Hz tones together. A "2" was composed of 700 Hz and 1,100 Hz together. A "3" was 700 Hz and 1,300 Hz.

I froze and grabbed Steve and nearly screamed in excitement that I'd found it. We both stared at the list, rushing with adrenaline. We kept saying things like, "Oh, shit!" and "Wow, this thing is for real!" I was practically shaking, with goose bumps and everything. It was such a Eureka moment. We couldn't stop talking all the way home. We were so excited. We knew we could build this thing. We now had the formula we needed! And definitely that article was for real.

That evening I went to Sunnyvale Electronics and bought some standard parts to build tone generators like the ones in the article. Immediately I found a tone generator kit and brought it back to Steve's house. Right there I soldered together two tone generators. Luckily, Steve had built a frequency counter before this, so we were able to put them together, plus a device that let you turn a dial and measure the tone it produced. For example, I could turn one dial and measure until I got the tone to approximately 700 Hz frequency. Then I could turn the other dial until it got to 900 Hz. Finally I'd play both tones at once and record the sound for about a second on a regular cassette tape recorder. Remember, those tones together meant a "1." Then I did the right combinations for the other digits. Eventually we had a seven-digit phone number and even a ten-digit number recorded.

Finally we set a tone to 2,600 Hz, which is the high E note that

supposedly seized the free line mentioned in the article. It worked!

After dialing a free 555 long-distance information number, we heard that chirp the article had talked about. Then we presumed that the phone system was waiting for tones to tell it where to connect to. But uh-oh. We played the tones from our tape recording but we weren't able to get the call to go through.

Man, it was so frustrating. No matter how hard we tried to get the frequencies right, they wavered. I just couldn't make them accurate. I kept trying, but I just couldn't perfect this thing. I realized I didn't have a good enough tone generator to prove the article true or false one way or the other.

But I was not about to give up.

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The next day was my first day at Berkeley. I was involved with my classes—I thought they were great classes—but I kept thinking about the Blue Box design. I took that *Esquire* article with me and started collecting every article from Sunday papers I could find about phone phreaks. I started posting articles I found about them on my dorm room wall. I started telling my friends what these phone phreaks were all about, how intelligent they must be, and how I was sure they were starting to take over the phone system all over the country.

So there I was at Berkeley, living in my little dorm room on the first floor of Norton Hall, my best school year ever. I could mesmerize an audience of kids with tales from this article and what Steve Jobs and I had been trying to do. I started gaining a reputation as the dorm's "phone phreak," which was fitting. Because one day I explored our dorm and found an unlocked telephone wire access box for our floor. I saw enough phone wires going up to the higher floors—there were a total of eight floors of dorm rooms, including my own above the common area—and I tapped pairs of wire and connected handsets to them. The idea was to

determine for a fact which lines were the ones going to which dorm rooms. So I ended up being able to play around and find any particular phone line I wanted to.

Even though I was usually shy and went unnoticed, suddenly this phone phreaking stuff brought me out to a position of prominence in my dorm, where everyone was seeking some sort of partying and fun.

It was around this time that I discovered another kind of phone phreak box, called a "Black Box." Instead of letting you dial free numbers, like the Blue Box did, the Black Box meant anyone who called you wouldn't be billed for the call.

I found the schematics for the Black Box in Abbie Hoffman's *Steal This Book*, an underground book I somehow managed to get at a regular bookstore. (They kept it under the counter so no one would follow the order in its title!)

That same year, an issue of *Ramparts* magazine came out with a really well-explained and completely illustrated article on how to build a Black Box with about two dollars' worth of parts from Radio Shack. All you needed were a capacitor, a resistor, and a switch or push button.

Here's how it worked: when someone called you long-distance you pushed the button to briefly tell the local phone company that you were answering. This connected the faraway line to your own. Because you didn't answer for two seconds minimum, the local phone company didn't send back a billing signal. Yet you were still connected to the caller, and the capacitor of the Black Box allowed their voice to reach your phone (and vice versa) without the phone company sensing any sort of connection. This device worked very well. In fact, one pole-vaulter in my dorm got a letter one day from his parents wondering why they hadn't been billed for two calls from Florida.

By the way, the phone company sued *Ramparts* after the article ran and drove it out of business by 1975.

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So while I'm playing with Black Boxes and spreading the word about the Blue Boxes, I started to seriously go to work on my own design. Only this time I tried a digital Blue Box design, which I knew would be able to produce precise, reliable tones. Looking back, I see that it was just a radical idea to do a digital Blue Box. In fact, I never saw or heard about another digital Blue Box. Making it digital meant I could make it extremely small and it was always going to work because it was based on a crystal clock to keep it accurate. That's the same way, by the way, watch crystals keep your watch running correctly.

I already had really good design skills by this point. I mean, I'd been designing and redesigning computers on paper all the way through high school and my last two years of college. I knew so much about circuit design, probably more than anyone I knew.

And then one day I did it. I designed my own digital Blue Box.

It was great. I swear to this day—the day I'm telling you this and the day you're reading it—I have never designed a circuit I was prouder of: a set of parts that could do three jobs at once instead of two. I still think it was incredible.

You see, the circuit, which generated codes corresponding to which button you pushed, used the chips in a very unusual way.

The way all electronics works, including chips, is that some signals are sent to the electronics, to their inputs. And the resulting signals come out of the chip on connections called outputs. Now, because I was familiar with the internal circuitry of the chips, I knew that tiny signals were actually being emitted by the inputs. After those tiny signals went through my button coding circuit, I directed them to a transistor amplifier, which supplied power to turn the chips on. So you can see the amazing thing. (At least, you can see it if you're an engineer.) The chips had to supply a signal to turn them on, and they did. That signal came from one side of the battery being connected, but not the other side.

I never have been able to do anything this out-of-the-box in any of my designs in my career at Hewlett-Packard or at Apple. That's saying a lot, because my designs have always been noted for being out-of-the-box. But this was cleverer than anything.

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Now, I won't say that getting a Blue Box working was just an instant thing. That's not what happens in engineering. I was in school, taking classes, and it was probably a couple of months before I did this design. But once I designed it, building it took a day.

I brought it to Steve's house, and we tried it on his phone there. It actually worked. Our first Blue Box call was to a number in Orange County, California—to a random stranger.

And Steve kept yelling, "We're calling from California! From California! With a Blue Box." He didn't realize that the 714 area code *was* California!

Instantly we got in my car and drove up from Steve's house to the dorm at Berkeley. We had promised our parents, who all knew about the project, that we would never do it from home. That one call to Orange County would be the only illegal one from either of our parents' houses.

I wanted to do the right thing. I didn't want to steal from the phone company—I wanted to do what the *Esquire* article said phone phreaks did: use their system to exploit flaws in the system. These days, phone phreaks shun those who do it to steal.

Also, I would've died to meet Captain Crunch, who was really the center of it all. Or any phone phreak; it just seemed so impossible that I'd ever meet anyone else with a Blue Box.

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One day Steve Jobs called me and said that Captain Crunch had actually done an interview on the Los Gatos radio station KTAO.

I said, "Oh my god, I wonder if there's any way to get in touch

with him.” Steve said he’d already left a message at the station but Captain Crunch hadn’t called back.

We knew we just had to get in touch with this most famous—infamous, really—and brilliant engineering criminal in the world. After all, he was the guy we’d been obsessed with for months; he was the guy we’d been reading about and telling stories about. We left messages at KTAO but never heard back from Captain Crunch. It looked like a dead end.

But then, the most coincidental thing happened. A friend of mine from high school, David Hurd, called me and wanted to catch up. When he came up to visit, I started to tell him all these incredible Captain Crunch stories and about the Blue Box and he said, “Well, don’t tell anyone, but I know who Captain Crunch is.” And I looked at him, floored. How could some random friend from high school know who Captain Crunch was?

I said, “What?”

“Oh yeah,” he said, “I know who he is. His real name is John Draper and he works at a radio station, KKUP in Cupertino.”

The next weekend, I was sitting with Steve at his house and told him what I’d found out. Steve immediately called the station and asked the guy who answered, “Is John Draper there?” He didn’t even say Captain Crunch.

But the guy said, “No, he dropped out of sight after the *Esquire* article.”

Hearing that, we knew we’d found the real Captain Crunch. We left our phone number with the guy just on the off chance that Captain Crunch might call us back. And in about five minutes, Captain Crunch actually called!

We picked up the phone, and he immediately told us who he was. But he said he didn’t want to say much on the phone. (I remember how, in the *Esquire* article, he had seemed pretty paranoid, sure that the phone line he was talking to the reporter on was bugged.)

Then we told him what kind of equipment we had, what we'd built. I told him it was a Blue Box that I myself had designed, and that it was digital. And he said, again, "Well, I can't talk on the phone about this, but I will come meet you in your dorm."

Man, I drove back to Berkeley just shaking the whole way. When I got there, I was telling everyone who'd listen, "Wow! Captain Crunch is coming here!" This guy I had made into a superhero—the hero of technology bandits or whatever you'd call him—the head guy, the best-known guy, was coming to my dorm room! And everybody was saying, "Can I come?"

But I said, no, no, I knew Captain Crunch wouldn't like that. So it was just my roommate, John Gott, Steve Jobs, and I who sat there in my dorm room, waiting and waiting.

Now, for some reason I was expecting this suave ladies' man to come through the door. I think it was because I'd read in the *Esquire* article that he'd tapped his girlfriend's phone line once and heard her talking to another guy, and then he called her up and said, "We're done." Just having a girlfriend, I guess, made him a ladies' man to me. I still had never even had a girlfriend.

But no. Captain Crunch comes to our door, and it turns out he's just this really weird-looking guy. Here, I thought, would be a guy who would look and act just far away and above any engineer in the world, but there he was: sloppy-looking, with his hair kind of hanging down one side. And he smelled like he hadn't taken a shower in two weeks, which turned out to be true. He was also missing a bunch of teeth. (Over the years, the joke I made up about him was that the reason he had no teeth was that he was stripping phone lines with them when the phone rang. Engineers know that the phone ringing signal is a high enough voltage to shock you really hard.)

So anyway, I saw him, and he didn't match my expectations. So I asked, "Are you Captain Crunch?" And he said, "I am he," and

he walked in just majestically. What a line that was. "I am he." And there he was.

He turned out to be this really strange, fun guy, just bubbling over with energy. And he's sitting on the bed, looking at all my phone phreak articles taped to my wall, and all the circuits and magazines, and also weird things like the twenty pounds of saltines I'd swiped from the cafeteria by stuffing a few packets in my pockets at every meal.

And he looked around and saw wires coming out of the telephones; I could tell he was surprised. I was sitting there thinking: Wow, this is the most amazing night of my life of all time, and it's just beginning!

He started talking to us. I noticed he was kind of like one of these very hyper people who keep changing topics and jumping around, talking about different times in his life and different stuff he did. I kept trying to impress him with my Blue Box. I boasted about how small it was, how few parts it took, and how it was digital—that was the main thing. I told him there was just one thing, I hadn't figured out how to make international calls yet. And he showed me the procedure right away. Strangely enough, it was the same procedure we'd read about in the *Esquire* article, but it didn't work then, don't ask me why.

Then suddenly Captain Crunch said, "Wait, wait a minute. I am going to go out to my car now and get my *automatic* Blue Box."

We knew right away that this was going to be some incredible piece of equipment that's going to be something special, like the digital Blue Box I'd designed. The way he said it—automatic. It was sort of a competitive thing.

I had this image of what this van must look like—with everything he needed to seize phone systems and other stuff in it. I imagined racks of engineering equipment and telephone equipment based on what I'd read in *Esquire*.

So I asked him, "Can I come?" I just had to see it. It was as if

I'd be seeing history, like one of the Seven Wonders of the World.

Well, I followed him out to the parking lot where his VW van was parked. It was completely empty. It was a totally empty van. All it had in it was this little Blue Box device on the floor, and a strange kind of crossbow-shaped thing, like a cross. It turned out that was the antenna he used to run San Jose Free Radio, a pirate radio station. He said the reason he ran it out of his van was so that no one from the FCC could ever pinpoint his location. Brilliant!

So that was to his credit, but still, all this equipment I expected to see wasn't there, and then there were his looks, and the strangely empty van. Everything was starting to not add up all of a sudden. I started to feel queasy and uncomfortable standing there. My previous ideas about what phone phreaks were supposed to be like were not meshing with the person I was looking at. This was a technical scallywag.

Then we went back to my dorm room and he took out this automatic box and showed me a few of its special abilities. The box had sliding switches—ten of them—to let you set up each of ten possible digits in a phone number. You could just push a button on the box—beep beep beep—and dial the whole number from there, no whistling or tone signals required! I was totally impressed by this, it was just really great.

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Later on, four of us—Steve Jobs, Captain Crunch, a guy named Alan McKittrick (who we called Groucho), and I—headed to Kips pizza parlor. We kept trading codes for dialing into various places and techniques like using Blue Boxes from pay phones.

At around midnight, we said our goodbyes. Captain Crunch wanted to go over to Groucho's house first and then drive his van home to where he lived in Los Gatos. So Steve Jobs and I took Steve's car back to his house in Los Altos, where my Pinto was parked.

Steve mentioned that his car had been having generator problems. When I asked what that meant, he said, "Pretty much the whole car can just suddenly shut down at some point."

About halfway down, in Hayward on what was then Highway 17, as I recall, it actually happened. The car lost all lights and power. Steve was able to pull over to the right near an exit and we walked to a gas station where there was a pay phone. We thought we'd call Groucho—a long-distance number from Hayward—and ask Draper to pick us up on his way down south.

Steve put a dime in the pay phone and dialed the operator. He told her he was about to make a "data call," to keep her from thinking that our line was going off the hook for the brief period while we used the Blue Box. He asked her to connect us to a free 800 number or long-distance directory assistance, some free call. Then we "blew it off" (seized the line with a 2,600 hertz tone) and Steve proceeded to use the Blue Box to call Groucho. But the operator came back on the line, so Steve hung up the phone quickly. This was not good!

We tried it again, telling the next operator that it was a data call and to ignore any weird light she might see. But the same thing happened. The operator came back on the line just before we made the connection. Again, Steve hung up instantly. We thought we were getting in deep trouble, that somehow our Blue Box had been detected.

Finally we decided to use coins and just call Groucho the legal way. We did and asked Captain Crunch to pick us up. All of a sudden, a cop pulled into the gas station and jumped out real fast. Steve was still holding the Blue Box when he jumped out, that's how fast it happened. We didn't even have time to hide it. We were sure that the operator had called the cops on us, and that this was the end for sure.

The cop was kind of heavysset and walked past me for some reason, shining his flashlight on the plants about eight feet in

front of me. I had long hair and a headband back then, so I guess he was looking for drugs we'd stashed. Then the cop started examining the bushes, rifling through them with his hands in the dark.

In the meantime, trembling with fear, Steve passed the Blue Box to me. He didn't have a jacket on, but I did. I slid it into my pocket.

But then the cop turned back to us and patted us down. He felt my Blue Box and I pulled it out of my pocket and showed it to him. We knew we were caught. The cop asked me what it was. I was not about to say, "Oh, this is a Blue Box for making free telephone calls." So for some reason I said it was an electronic music synthesizer. The Moog synthesizer actually had just come out, so this was a good phrase to use. I pushed a couple of the Blue Box buttons to demonstrate the tones. This was pretty rare, as even touch-tone phones were still kind of rare in this part of the country then.

The cop then asked what the orange button was for. (It was actually the button that sounded the nice pure 2,600 Hz tone to seize a phone line.) Steve told the cop that the orange button was for "calibration." Ha!

A second cop approached. I guess he had stayed back in the police car at first. He took the Blue Box from the first cop. This device was clearly their point of interest, and surely they knew what it was, having been called by the phone operator. The second cop asked what it was. I said it was an electronic music synthesizer. He also asked what the orange button was for, and Steve again said that it was for calibration. We were two scared young cold and shivering boys by this time. Well, at least Steve was shivering. I had a coat.

The second cop was looking at the Blue Box from all angles. He asked how it worked and Steve said that it was computer-controlled. He looked at it some more, from every angle, and

asked where the computer plugged in. Steve said that “it connected inside.”

We both knew the cops were playing with us.

The cops asked what we were doing and we told them our car had broken down on the freeway. They asked where it was and we pointed. The cops, still holding the Blue Box, told us to get in the backseat of their car to go check out the car story. In the

How Ma Bell Helped Us Build the Blue Box

In 1955, the *Bell System Technical Journal* published an article entitled “In Band Signal Frequency Signaling” which described the process used for routing telephone calls over trunk lines with the signaling system at the time. It included all the information you’d need to build an interoffice telephone system, but it didn’t include the MF (multifrequency) tones you needed for accessing the system and dialing.

But nine years later, in 1964, Bell revealed the other half of the equation, publishing the frequencies used for the digits needed for the actual routing codes.

Now, anybody who wanted to get around Ma Bell was set. The formula was there for the taking. All you needed were these two bits of information found in these two articles. If you could build the equipment to emit the frequencies needed, you could make your own free calls, skipping Ma Bell’s billing and monitoring system completely.

Famous “phone phreaks” of the early 1970s include Joe Engressia (a.k.a. Joybubbles), who was able to whistle (with his mouth) the high E tone needed to take over the line. John Draper (a.k.a. Captain Crunch) did the same with the free whistle that came inside boxes of Cap’n Crunch. A whole subculture was born. Eventually Steve Jobs (a.k.a. Oaf Tobar) and I (a.k.a. Berkeley Blue) joined the group, making and selling our own versions of the Blue Boxes. We actually made some good money at this.

backseat of a cop car you know where you are going eventually: to jail.

The cops got in the front. I was seated behind the driver. The cop in the passenger seat had the Blue Box. Just before the car started moving, or maybe just after, he turned to me and passed me the Blue Box, commenting, "A guy named Moog beat you to it."

Chapter 7

Escapades with Steve

The internal joy I felt when the cop believed our story about the Blue Box being the Moog synthesizer is almost indescribable.

Not only were we not being arrested for making illegal calls with or owning a Blue Box, but these supposedly intelligent cops had totally bought our B.S. God, I wanted to laugh out loud. Our moods changed instantly.

I mean, one second we thought we were being driven off to jail, and the next we realized we had bamboozled the police. Bamboozled up the yazoo! This was such an important lesson to learn in life, and such a continuing theme for me. Some people will just believe the strangest stuff, stuff that doesn't bear any semblance to reality.

After the cops dropped us off, we waited at the gas station until finally Captain Crunch showed up in his van. That van sure was scary to ride in as a passenger. It felt like it was going to fall over, it was so rickety. It didn't feel solid at all. It was probably about 2 a.m. by the time we got back to Steve's house all the way down in Los Altos. I picked up my car—I had an ochre-colored Pinto at this time—and drove it back to Berkeley.

I was tired. I shouldn't have been on the road. Because you know what happened? Somewhere near Oakland on Highway 17,

I fell asleep at the wheel. I don't know how long my eyes were closed, but suddenly I opened my eyes and it looked to me like the guardrail was jumping onto my windshield. It looked so strange, like a dream. I grabbed the steering wheel, yanked it as hard as I could to the right, and the car just started spinning around and around.

The only thing holding me in that car was the seat belt.

As the car was spinning, I was thinking, This is it. I might die. I could really die. But then the car slid to a stop up against the center guardrail, and it turned out that only the side of the Pinto that hit the guardrail was damaged. But it totaled my car.

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Losing my Pinto changed my life completely. One of the major parts of my life at Berkeley was taking groups of people down to Southern California or even as far south as Tijuana, Mexico, on weekends. Actually, my first thought after the crash wasn't, Oh, thank god I'm alive, but Man, now I'm not going to be able to take my friends on wild adventures anymore.

The car crash was the main reason that, after this school year, my third year at Berkeley, I went back to work instead of coming back to school. I needed to earn money, not just for the fourth year of college but also for a new car.

If I hadn't gotten in the car accident that year, I wouldn't have quit school and I might never have started Apple. It's weird how things happen.

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But for the rest of the year at Berkeley, I kept playing with my Blue Box. Captain Crunch's design had given me an idea: to add a single little button where I could preprogram a ten-digit number.

The number I chose to dial was this weird joke line in Los Angeles. It was called Happy Ben. When you called it, this cranky old guy—he sounded like a real old guy—would answer in this

old voice like gravel: "Hey," he'd say, "it's me, Happy Ben." And then he'd sing, off-key, and with no music: "Happy days are here again / happy days are here again / happy days are here again / happy days are here again." And then, "Yep, it's me again. It's Ben."

Don't ask me why, but of all the joke lines in the world I now had free access to with my Blue Box, that one number always cheered me up and made me smile. It was just the fact that this grumpy-sounding old guy would sing that song in such a truly happy way. Somehow that style of humor made me laugh. I hope to do the same thing myself someday. Maybe I can sing the national anthem on a joke line. I still might.

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Now that I had a Blue Box that could call anywhere, even internationally, I had a lot of fun calling joke lines all over the world. I'd walk up to a pay phone, dial some 800 number, seize the line with the Blue Box, push the automatic button—beep beep beep—and there he was again. Happy Ben singing "Happy Days Are Here Again." It was my favorite thing.

But I hadn't forgotten what was supposed to be the real mission of phone phreaking: not to mess up the system, but to find flaws and curious things and secrets the phone company never told anyone about. And I really did stick with the honesty thing. Even when I made my calls to friends, relatives, to people I normally would've called anyway, I made a point of paying for those calls. I didn't use the Blue Box. To me, that would have been stealing, and that wasn't what I was about.

But I did like to use the Blue Box to see how far it could get me. For instance, I would make a call to an operator and pretend I was a New York operator trying to extend the lines for phase measurements, and she would connect me to London. Then I'd talk that operator into connecting me to Tokyo. I would go around the world like this sometimes three times or more.

And by this time I got great at sounding official, or doing accents, all to fool operators around the world. I remember one very, very late night in the dorm when I decided to call the pope. Why the pope? I don't know. Why not? So I started by using the Blue Box to call Italy Inward (country code 121), then I asked for Rome Inward, and then I got to the Vatican and in this heavy accent I announced I was Henry Kissinger calling on behalf of President Nixon. I said, "Ve are at de summit meeting in Moscow, and ve need to talk to de pope."

And a woman said, "It's five-thirty here. The pope is sleeping." She put me on hold then for a while, and then told me they were sending someone to wake him and asked if I could call back. I said yes, in an hour.

Well, an hour later I called back and she said, "Okay, we will put the bishop on, who will be the translator." So I told him, still in that heavy accent, "Dees is Mr. Kissinger." And he said, "Listen, I just spoke to Mr. Kissinger an hour ago." You see, they had checked out my story and had called the real Kissinger in Moscow.

Ha! But I didn't hang up. I said, "You can verify my number. You can call me back." And I gave him a U.S. number that would call a loop-back number so they wouldn't find out my number. But they never called back, which was too bad.

Years later, though, I couldn't stop laughing when I saw an article about me where they were interviewing Captain Crunch. He said I was calling the pope to make a confession!

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For ages and ages, I always told people how I was the ethical phone phreak who always paid for my own calls and was just exploring the system. And that was true. I used to get huge phone bills, even though I had my Blue Box that would've let me make any call for free.

But one day Steve Jobs came along and said, "Hey, let's sell

these.” So by selling them to others we really were getting the technology out to people who were using it to call their girlfriends and the like and save money on phone calls. So looking back, I guess that, yes, I aided and abetted that crime.

We had a pretty interesting way of selling them. What we would do is Steve and I would find groups of people in various dorms at Berkeley to sell them to. I was always the ring-leader, which was really unusual for me. I was the one who did all the talking. You know, I thought I’d be so famous doing this, but it’s funny, I didn’t know you had to talk to a reporter to get your phone phreak handle (mine was Berkeley Blue) in articles.

Anyway, the way we did it was just by knocking on doors. How do you know you’re not walking up to somebody who’s going to turn you in? Someone who might see it as a crime? Well, we’d knock on a door (usually a door in a male dorm) and ask for someone nonexistent like, “Is Charlie Johnson there?” And they’d say, “Who’s Charlie Johnson?”

And I’d say, “You know, the guy that makes all the free phone calls.” If they sort of seemed cool—and you could tell by their face if they wanted to talk about such a thing as illegal free phone calls—I’d add, “You know, he has the Blue Boxes?”

Sometimes they might say, “Oh my god, I’ve heard about those things.” And if they sounded really cool enough, and every once in a while they did, then one of us pulled a Blue Box out of our pockets. They’d say something like, “Wow! Is that what they look like? Is that real?”

And that’s how we knew we had the right guy and he wouldn’t turn us in. Then one of us would say, “Tell you what, we’ll come back at 7 p.m. tonight; have everyone you know who knows someone in a foreign country here and we’ll give you a demo.”

And we’d come back at 7 p.m. We’d run a wire across their dorm room and we’d hook it up to the tape recorder. That way,

everything was tape-recorded—every single sale we ever did was tape-recorded. Just to play it safe.

We made a little money selling Blue Boxes. It was enough at the time. Originally I would buy the parts to hand-build one for \$80. The distributor in Mountain View where I got the chips (no electronics stores sold chips) charged a ton for small quantities. We eventually made a printed circuit board and, making ten or twenty at a time, got the cost down to maybe \$40. We sold them for \$150 and split the revenue.

So it was a pretty good business proposition except for one thing. Blue Boxes were illegal, and we were always worried about getting caught.

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One time Steve and I had a Blue Box ready for sale. Steve needed some extra money, and he really wanted us to sell the box that day. It was a Sunday. Before driving up to Berkeley to sell the Blue Box, we stopped to eat at a Sunnyvale pizza parlor. While eating our pizza, we noticed a few guys at the next table. They looked cool, and we started talking to them. It turned out they were interested in seeing one and buying it.

We then went to a rear hallway of the pizza parlor, where there was a pay phone. Steve pulled out the Blue Box. They gave us a number in Chicago, in the 312 area code, to test it. The call went through to a ringing phone, which no one answered.

The three guys were really excited and told us they wanted the Blue Box but couldn't afford it. Steve and I headed out to the parking lot to get into Steve's car. And just very quickly, before Steve started the engine, one of the guys popped up next to the driver's-side window with a big long black gun barrel pointed right at us.

He demanded the Blue Box.

Steve nervously handed it to him. And the thieves went to their own car. As we sat there, stunned, an amazing thing hap-

pened. One of the guys came back to the car and explained he didn't have the money yet, but he did want the Blue Box. And that they would pay us eventually. And he wrote down a phone number and a name for us to call him at. His name was Charles.

After a few days, Steve called the number. Someone answered, and when we asked for Charles, he gave us the number of a pay phone. We knew it was a pay phone because back then, if the last four digits of any number started with a 9 or a 99, it was certainly a pay phone.

Steve called that number, and Charles answered. He said he would pay us eventually for the Blue Box, but he needed to know how to use it.

Steve tried to talk him into returning it to us. Charles said he wanted to meet us somewhere. We were too scared to meet him, even in a public place. I came up with the idea of telling him a method to use that would get him billed for every call—like, to start your call by dialing an 808 number, which is an area code for Hawaii. I also thought of telling him a way to use it that would get him caught. Like dialing 555 information calls, which look suspicious when they last for hours.

If only I'd been more of a joker, I would've thought to tell him to start by dialing the number of a police station.

But I didn't recommend any of those things, and in the end, Steve hung up. We were too scared to do anything, and for sure Charles and those guys never learned how to use it.

Chapter 8

HP and Moonlighting as a Crazy Polack

This much I know for sure: I was meant to be an engineer who designs computers, an engineer who writes software, an engineer who tells jokes, and an engineer who teaches other people things.

Now, finally, there was a time in my life—a time right after that third year at Berkeley—that I finally got my dream job. But it wasn't a job building computers. It was a job designing calculators at Hewlett-Packard. And I really thought I would spend the rest of my life there. That place was just the most perfect company.

This was January of 1973, and for an engineer like me, there was no better place to work in the world. Unlike a lot of technology companies, Hewlett-Packard wasn't totally run by marketing people. It really respected its engineers. And that made sense, because this was a company that had made engineering tools for years—meters, oscilloscopes, power supplies, testers of all types, even medical equipment. It made all the things engineers actually used, and it was a company driven by engineers on the inside so far as what engineers on the outside needed. Man, I loved that.

For just a few months before that, right after I finished Berke-

ley in June, I worked at a much, much smaller company, called Electroglas. That was a blast, too. Getting that job was almost too easy. I'd looked in the newspaper ads, and the first ad I saw was for an electronics technician for \$600 a month, or close to that. I called them up and they said, "Come on down for an interview." Well, I went down and they gave me this incredibly easy written test—you know, with electronic formulas and all. Of course, I knew that stuff. I'd known it forever. They interviewed me and instantly hired me, so I had a job. And they paid me enough that I was actually able to get my first apartment. It was in Cupertino, just a mile from my parents' house. And it was just the greatest, greatest thing.

But six months later I heard from my old friend Allen Baum, who by then was working as an intern at Hewlett-Packard. He was excited, telling me he was actually hanging around the guys who'd designed the HP 35 calculator. This, to me, was the most incredible invention of all time.

I'd been a slide rule whiz in high school, so when I saw the calculator, it was just amazing. A slide rule was kind of like a ruler—you had to look at it precisely to read the values. The most accurate number you could get was only three digits long, however, and even that result was always questionable. With a calculator, you could punch in precisely the digits you wanted. You didn't have to line up a slider. You could type in your numbers exactly, hit a button, and get an answer immediately. You could get that number all the way out to ten digits. For example, the real answer might be 3.158723623. An answer like that was much more precise than anything engineers had ever gotten before.

Well, the HP 35 was the first scientific calculator, and it was the first in history that you could actually hold in your hand. It could calculate sines and cosines and tangents, all the trigonometric and exponential/logarithmic functions engineers use to calculate and to do their jobs. This was 1973, and back then cal-